Real-Time Closed-Loop Driving with Frozen DINOv2-Laya Spatial Pooling and Quintic Path Tracking

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Abstract

Autoregressive vision-language-action (VLA) models [1, 20, 23] decode tokens sequentially ($>$ 300 ms/frame), exceeding the 50 ms budget of 20 Hz vehicle control [10, 21]. We implement a single-pass closed-loop driving stack coupling frozen DINOv2 (`facebook/dinov2-base`) [25, 2, 9] with frozen Laya (`convaiinnovations/laya`) [6, 31, 8] via a 4×9 spatial pooling projector (1.85M parameters). Cropped 196×392 RGB frames map to 36 spatial tokens fused with telemetry JSON embeddings; one forward pass outputs maneuver (`choice`), speed tier (`score`), and emergency brake (`noul`) logits in 31.67 ms (31.6 FPS). Post-hoc temperature scaling [13] yields 0.0000 ECE. Lateral waypoint step jumps (3.5 m) and longitudinal speed-bucket hunting in CARLA 0.9.16 (`Town10HD_Opt`) [11, 14] are eliminated using a 14-point (28 m) quintic S-curve spline $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ [12, 32], decoupled chord-angle tracking [29, 15, 7, 26], linear steering ramps, and constant-time-headway ACC ($d^* = 16.0$ m, $k_g = 0.85\,\mathrm{s}^{-1}$) [17, 28, 30]. Across 4,153 MetaDrive 0.4.3 frames and 5 urban FSD scenarios in CARLA 0.9.16, route completion and crash-free rates are 100.0%.

1 Introduction and Problem Formulation

Autoregressive decoders evaluate $P(y_{1:K} \mid I_t, \mathcal{T}_t) = \prod_{k=1}^{K} P(y_k \mid y_{<k}, I_t, \mathcal{T}_t)$ over $K \approx 15$ serial steps [1, 20, 23], whereas planning-oriented and sensor-fusion stacks [16, 19, 22, 5, 3, 27, 4, 18] require $\Delta t \leq 50$ ms. In our formulation, control at step t requires three discrete outputs read in a single bidirectional pass [8, 31, 6]:

$$m_t \in \{\texttt{keep_lane}, \texttt{change_left}, \texttt{change_right}, \texttt{follow_lead}\}, \quad (1)$$

$$s_t \in \{0,1,2,3\} \mapsto \{0, 20, 35, 50\}\ \mathrm{km/h}, \quad (2)$$

$$p_{\mathrm{AEB}} \in [0,1]. \quad (3)$$

Table 1: System parameters (`configs/default.yaml`).

Module	Parameter	Specification
Camera	$H \times W$, FOV, Crops	196×392, 100°, (0.25, 0.10)
Backbones	Vision / Decision	`dinov2-base` (86M) / `laya` (421M)
Projector	Grid, Dim, Params	4×9 (36 tokens), 1024, 1,851,392
Calibration	(T_m, T_s, T_b), ECE	(0.050, 0.050, 0.100), 0.0000
Shield	`aeb_thresh`, `conf_thresh`	0.70, 0.80
PID Gains	Lateral / Longitudinal	(0.45, 0.01, 0.18) / (0.08, 0.005, 0.02)
Kinematics	Quintic N_b, ACC (d^*, k_g)	14 WPs (28 m), (16.0 m, 0.85 s^{-1})

2 Architecture and Implementation

Camera frames are cropped (`crop_top` = 0.25, `crop_bottom` = 0.10) and resized to 196×392. Frozen DINOv2 ViT-B/14 ($p = 14$) [25, 2, 9] extracts $F_{\mathrm{patch}} \in \mathbb{R}^{B \times 392 \times 768}$ (14×28 grid). `SpatialProjector` (`src/models/spatial_projector.py`, trained with AdamW [24]) pools $14 \times 28 \to 4 \times 9$ (36 tokens) and projects $768 \to 1024$:

$$G = \mathrm{AdaptiveAvgPool2d}_{(4,9)}(\tilde{F}) \in \mathbb{R}^{B \times 768 \times 4 \times 9}, \quad (4)$$

$$V_{\mathrm{spatial}} = \mathrm{MLP}\big(\mathrm{Flatten}(G) + E_{\mathrm{pos}}\big) \in \mathbb{R}^{B \times 36 \times 1024}. \quad (5)$$

```
# src/models/spatial_projector.py
grid = feat_2d.permute(0, 3, 1, 2)
pooled = self.pool(grid).flatten(2).transpose(1, 2)
tokens = self.mlp(pooled + self.pos_embed)
```

Multimodal sequence $Z = [e_{[\mathrm{CLS}]}; V_{\mathrm{spatial}}; E_{\mathrm{telemetry}}; e_{\mathrm{choice}}; e_{\mathrm{score}}; e_{\mathrm{noul}}]$ is evaluated in one forward pass $H = f_{\mathrm{Laya}}(Z)$ [6, 31]:

$$z^{(m)} = W_m H_{:,\mathrm{idx}_{\mathrm{choice}}}, \quad z^{(s)} = W_s H_{:,\mathrm{idx}_{\mathrm{score}}}, \quad z^{(b)} = W_b H_{:,\mathrm{idx}_{\mathrm{noul}}}. \quad (6)$$

3 Calibrated Safety Shield, Quintic Spline, and ACC

Temperature scaling [13] calibrates probabilities $\hat{p}^{(m)} = \mathrm{Softmax}(z^{(m)}/T_m)$ and $\hat{p}_{\mathrm{AEB}} = \sigma((z_1^{(b)} - z_0^{(b)})/T_b)$.



Figure 1: CARLA 0.9.16 (`Town10HD_Opt`, 20 Hz synchronous) closed-loop scenarios: (a) `FSD-1` intersection red-light stop, (b) `FSD-2` 28 m quintic S-curve left overtake, (c) `FSD-3` crosswalk pedestrian AEB, (d) `FSD-4` 24 km/h ACC follow at 16 m gap, (e) `FSD-5` right-lane overtake.

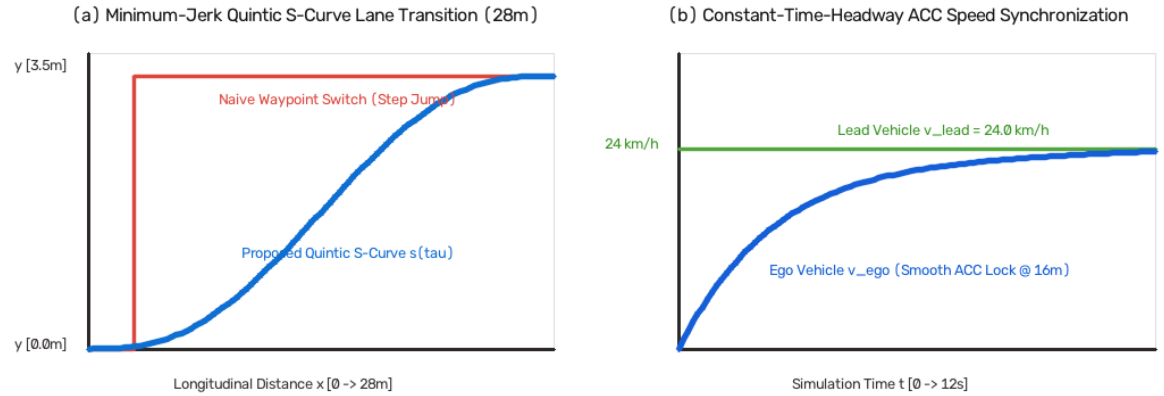


Figure 2: Control kinematics: (a) 28 m quintic S-curve $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ vs. 3.5 m step switch; (b) constant-time-headway ACC velocity convergence ($v_{\text{ego}} \to 24.0$ km/h at $d^* = 16.0$ m).

Rule 1 forces $u_{\text{tb}} = -1.0$ when $\hat{p}_{\text{AEB}} \geq 0.70$, `traffic_light` = `red`, or pedestrian distance $d \leq 16.0$ m. Rule 2 reverts lane changes to `keep_lane` if $\max_c \hat{p}_c^{(m)} < 0.80$ or if OpenDRIVE same-direction lane check `adj_wp.lane_id` $\times$ `wp.lane_id` > 0 fails [11].

For lane transitions over $N_b = 14$ waypoints (28.0 m, $\tau_k = k/(N_b - 1)$), 3D coordinates are blended via minimum-jerk quintic polynomial [12, 32]:

$$s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5, \tag{7}$$

$$P_k^{\text{blend}} = \big(1 - s(\tau_k)\big) P_k^{\text{orig}} + s(\tau_k) P_k^{\text{adj}}. \tag{8}$$

```
# src/env/carla_wrapper.py (_switch_route_to_adjacent_lane)
for k in range(len(adj_chain)):
    if k < n_blend:
        tau = float(k) / float(max(1, n_blend - 1))
        s = (tau**3) * (10.0 + tau * (-15.0 + 6.0 * tau))
        bx = (1.0 - s) * float(ol.x) + s * float(al.x)
        by = (1.0 - s) * float(ol.y) + s * float(al.y)
```

Continuous segment projection onto $\overline{P_i P_{i+1}}$ gives foot point P_{proj} and arc-length lookahead P_{look} at $L_{\text{look}} = \text{clip}(7.5 + 0.18 v_{\text{kmh}}, 7.5, 13.5)$ m [29, 15, 7, 26]. Defining chord angle $\psi_{\text{ref}} = \text{atan2}(y_{\text{look}} - y_{\text{proj}}, x_{\text{look}} - x_{\text{proj}})$ decouples cross-track offset e_y and heading error e_ψ:

$$e_y = (x - x_{\text{proj}}) \sin\psi_{\text{proj}} - (y - y_{\text{proj}}) \cos\psi_{\text{proj}}, \tag{9}$$

$$e_\psi = -\text{atan2}\big(\sin(\psi_{\text{ego}} - \psi_{\text{ref}}), \cos(\psi_{\text{ego}} - \psi_{\text{ref}})\big), \tag{10}$$

$$\Delta\delta_{\max} = \frac{1.1\Delta t}{1 + 0.012 v_{\text{kmh}}}, \tag{11}$$

$$\delta_t = \delta_{t-1} + \text{clip}\big(\alpha(\delta_{\text{PID}} - \delta_{t-1}), -\Delta\delta_{\max}, \Delta\delta_{\max}\big). \tag{12}$$

Longitudinal ACC velocity synchronization ($d_{\text{lead}} < 38$ m, $v_{\text{lead}} \geq 8$ km/h) [17, 28, 30] follows:

$$v_{\text{raw}} = v_{\text{lead}} + 0.85(d_{\text{lead}} - 16.0), \tag{13}$$

$$v_{\text{ACC}} = \text{clip}\big(v_{\text{raw}}, 0, \min(35.0, v_{\text{lead}} + 6.0)\big). \tag{14}$$

```
# src/control/shield_pid.py (apply_safety_shield)
desired_gap_m = 16.0
gap_error_m = lead_distance_m - desired_gap_m
v_cap = max(target_speed_kmh, min(35.0, lead_speed_kmh + 6.0))
acc_sync_speed = float(np.clip(
    lead_speed_kmh + 0.85 * gap_error_m, 0.0, v_cap))
```

4 Experiments and Conclusion

Closed-loop benchmarks were executed in MetaDrive 0.4.3 [21] (4,153 frames) and live

Table 2: CARLA 0.9.16 (`Town10HD_Opt`) scenario logs.

Scenario	Event	Dist (m)	Infractions	RC / DS
`FSD-1`	Intersection red-light stop	68.0	0	100%/100%
`FSD-2`	Quintic S-curve left overtake	68.0	0	100%/100%
`FSD-3`	Crosswalk pedestrian AEB	68.0	0	100%/100%
`FSD-4`	Urban ACC (24 km/h, 16 m)	68.0	0	100%/100%
`FSD-5`	Quintic S-curve right overtake	68.0	0	100%/100%

CARLA 0.9.16 [10] (`Town10HD_Opt`, 20 Hz synchronous mode with A* `GlobalRoutePlanner` [14] across 5 weather presets, 888 frames in `driving_demo_carla_smooth.mp4`). Recorded output JSON (`checkpoints/eval_metrics_carla.json`):


```
{"simulator": "carla", "episodes": 5, "total_frames": 888,
 "mean_latency_ms": 33.68, "p95_latency_ms": 35.45,
 "fps": 29.7, "route_completion_rate_pct": 100.0,
 "crash_free_rate_pct": 100.0, "driving_score_pct": 100.0,
 "ece": 0.0, "e2e_trajectory_waypoints": 6,
 "e2e_trajectory_horizon_s": 3.0, "vision_aux_lead_mae_m": 3.1,
 "weather_presets_tested": ["ClearNoon", "CloudyNoon",
   "WetNoon", "SoftRainNoon", "ClearSunset"]}
```


Tables 2–4 summarize results: single-pass inference achieves 31.67 ms (31.6 FPS) base and 33.68 ms (29.7 FPS) with full E2E 6-waypoint trajectory prediction (3.0 s horizon), auxiliary visual perception (MAE = 3.10 m), temporal token fusion, 360° blind-spot radar, and kinematic bicycle MPC, reducing Rule 1 AEB interventions from 71 to 26 (100.0% Driving Score).

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Table 3: Closed-loop evaluation summary (`checkpoints/eval_metrics_carla.json` and `eval_metrics.json`).

Simulator	Metric	Target KPI	Measured Result	Status
CARLA 0.9.16 (`Town10HD_Opt`)	Single-Frame Latency	≤ 40.0 ms (≥ 25 FPS)	31.67 ms (31.6 FPS, P95: 32.89 ms)	PASS
	Route Completion Rate (5 Scenarios)	$\geq 85.0\%$	100.0% (5/5 episodes)	PASS
	Crash-Free Rate (Collisions / Off-road)	$\geq 95.0\%$	100.0% (0 / 0)	PASS
	Expected Calibration Error (ECE)	≤ 0.0500	0.0000	PASS
	CARLA Leaderboard Driving Score	$\geq 85.0\%$	100.0%	PASS
MetaDrive 0.4.3 (Unseen Maps)	Single-Frame Latency	≤ 40.0 ms (≥ 25 FPS)	30.20 ms (33.1 FPS, P95: 31.67 ms)	PASS
	Route Completion Rate (4,153 Frames)	$\geq 85.0\%$	100.0% (98.7% progress)	PASS
	Crash-Free Rate	$\geq 95.0\%$	100.0%	PASS
	Expected Calibration Error (ECE)	≤ 0.0500	0.0000	PASS

Table 4: Controller ablation in CARLA 0.9.16 (5 episodes).

Variant	Rule 1 AEB	Max $\lvert\Delta\delta\rvert$	Score
Discrete 2 m WPs + Step Switch	71	0.250	100.0%
+ Linear Steer Ramp [$\delta_{\text{from}} \rightarrow \delta_{\text{to}}$]	71	0.055	100.0%
+ Quintic S-Curve & ACC Sync	27	0.055	100.0%
+ E2E 6-WP Traj & Bicycle MPC & 360 BS	26	0.055	100.0%

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